

CSC 112: Computer Operating Systems

Lecture 5

Scheduling

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CPU/IO Bursts

- A typical task alternates between bursts of CPU and I/O
 - It uses the CPU for some period of time, then does I/O, then uses CPU again (A job may be pre-empted and forced to give up CPU before finishing current CPU burst)

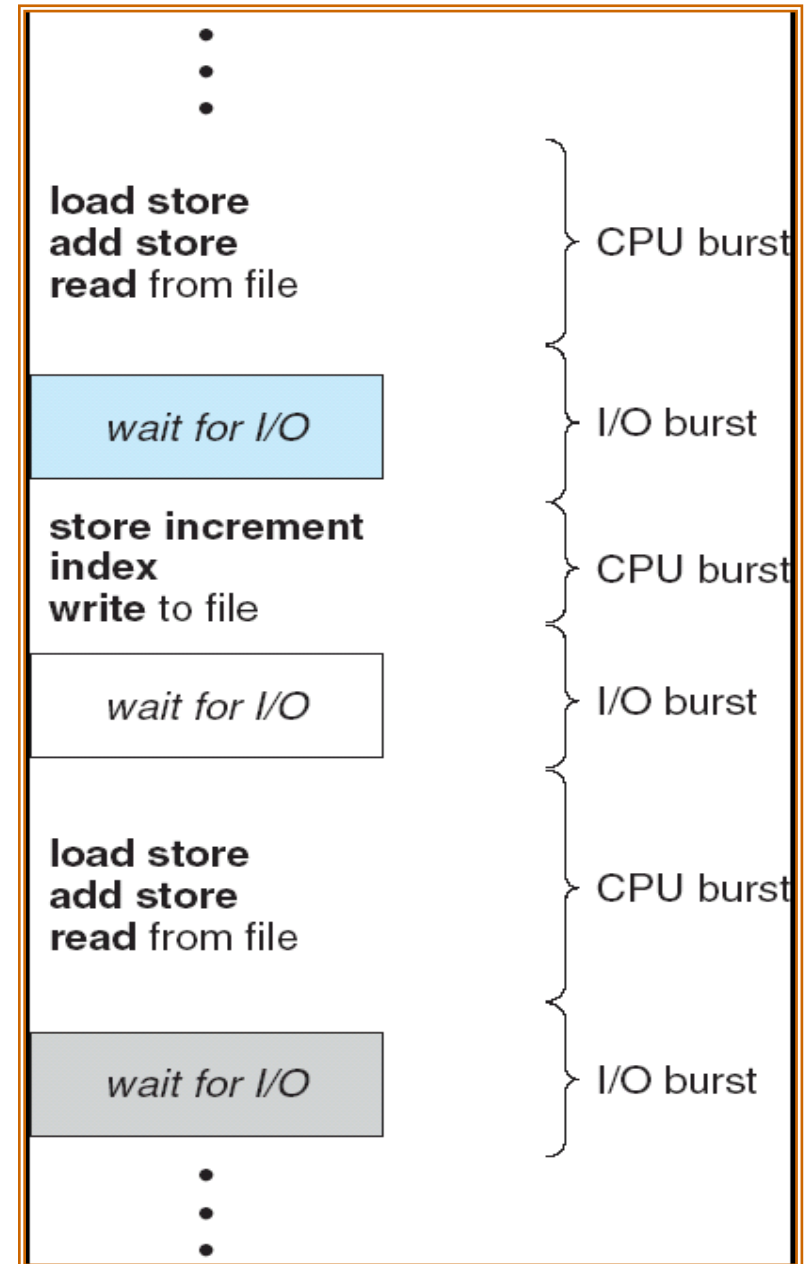
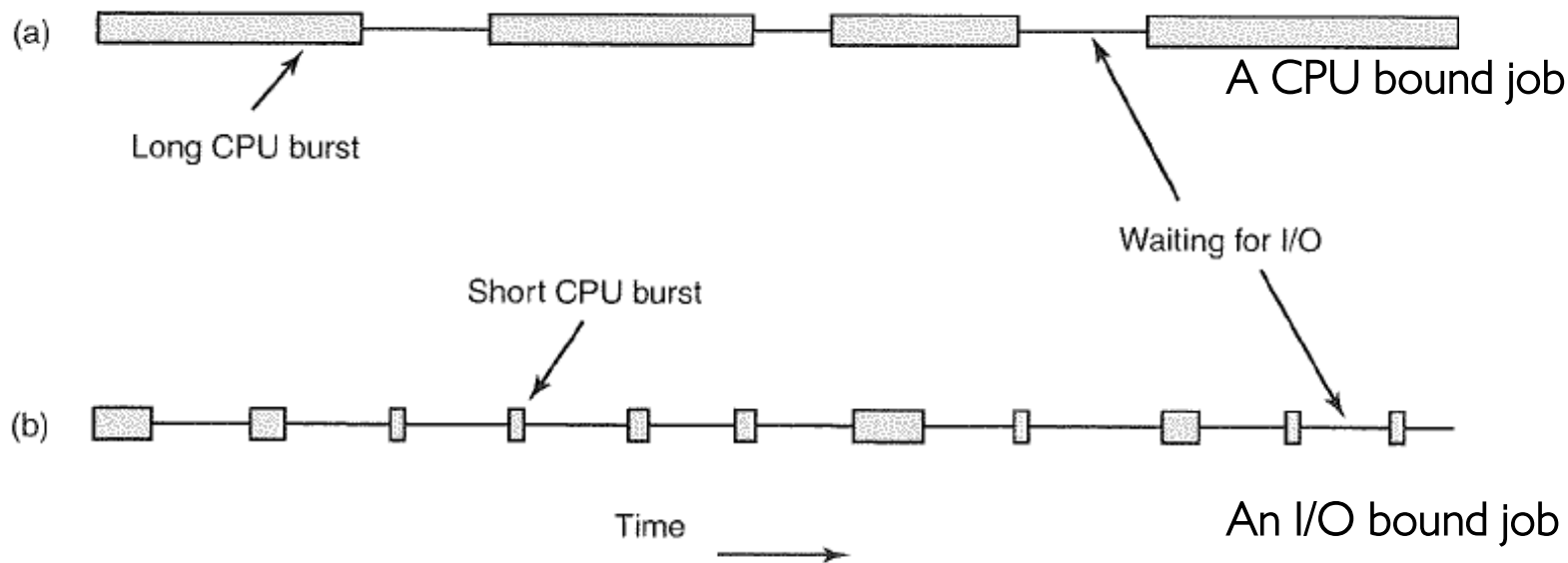
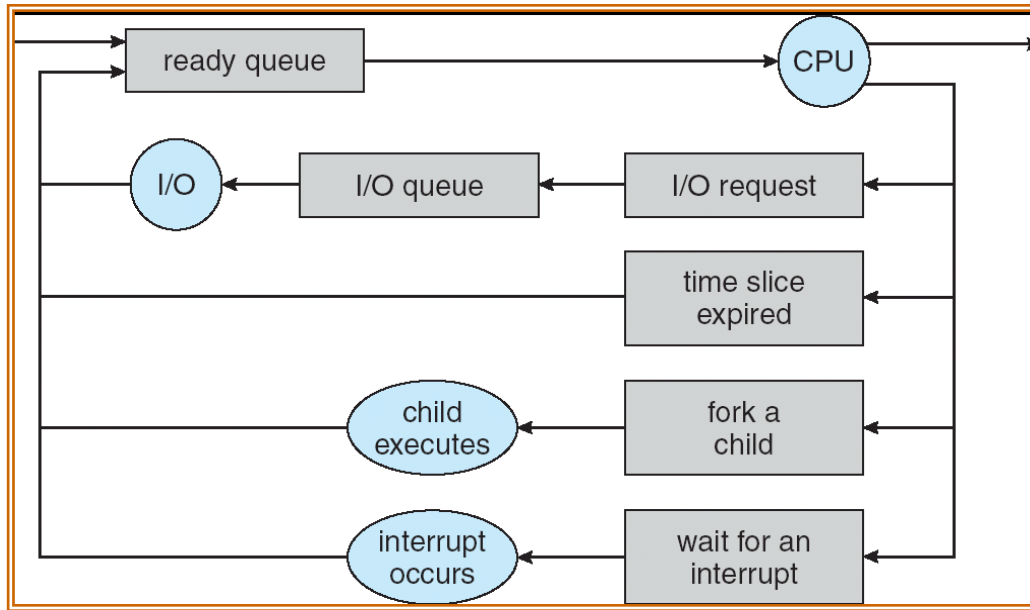


Figure 2-38. Bursts of CPU usage alternate with periods of waiting for I/O. (a) A CPU-bound process. (b) An I/O-bound process.

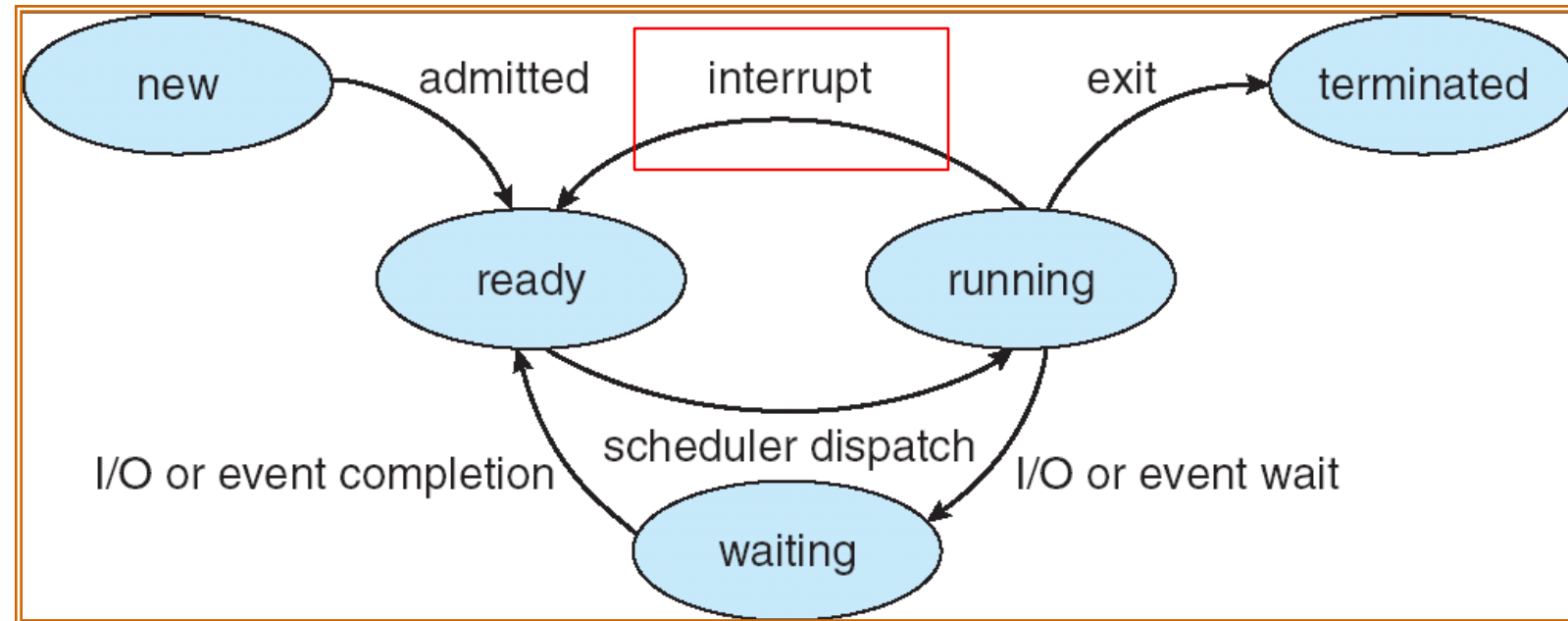
The Scheduling Problem



- **Scheduling:** When multiple jobs are ready, the scheduling algorithm decides which one is given access to the CPU
 - We use the term “task” to refer to a runnable entity in the OS, which may be a process or a thread. We use the term “job” to refer to a CPU burst of a task

Preemptive vs. Non-Preemptive Scheduling

- With non-preemptive scheduling, once the CPU has been allocated to a process, it keeps the CPU until it releases the CPU either by terminating or by blocking for IO.
- With preemptive scheduling, the OS can forcibly remove a process from the CPU without its cooperation
- Transition from “running” to “ready” only exists for preemptive scheduling



Performance Metrics

- **Response time**: the total time taken for a job to complete its execution, starting from the moment it arrives until it finishes. It includes all phases of the process lifecycle: waiting in queues, execution on the CPU, and any I/O operations. It can be calculated as $\text{CompletionTime} - \text{ArrivalTime}$.
 - Also called turn-around time
- **Initial waiting time**: the time a job spends waiting in the ready queue before it gets its first chance to execute on the CPU
- **CPU utilization**: percent of time when CPU is busy
- **Throughput**: # of jobs that complete their execution per time unit
- Different systems may have different requirements
 - Maximize CPU utilization
 - Maximize Throughput
 - Minimize Average Response time
 - Minimize Average Waiting time
 - Typically, these goals cannot be achieved simultaneously by a single scheduling algorithm

Common Scheduling Algorithms

- First-Come-First-Served (FCFS) Scheduling
- Round-Robin (RR) Scheduling
- Shortest-Job-First (SJF) Scheduling
- Priority-Based Scheduling
- Multilevel Queue Scheduling
- Multilevel Feedback-Queue Scheduling

First-Come, First-Served (FCFS) Scheduling

- First-Come, First-Served (FCFS)
 - Also “First In, First Out” (FIFO) or “Run until done”

- Example:

<u>job</u>	<u>Burst Time</u>
P_1	24
P_2	3
P_3	3

- Suppose jobs arrive in the order of P_1, P_2, P_3 at time 0, i.e., P_1 arrives at time 0, P_2 arrives at time ϵ , P_3 arrives at time 2ϵ
The Gantt Chart for the schedule is:



- Initial waiting times: $P_1: 0$; $P_2: 24$; $P_3: 27$
 - Response times: $P_1: 24$; $P_2: 27$; $P_3: 30$
 - Average initial waiting time: $(0 + 24 + 27)/3 = 17$
 - Average response time: $(24 + 27 + 30)/3 = 27$
- **Convoy effect:** short job stuck behind long job



FCFS Scheduling (Cont.)

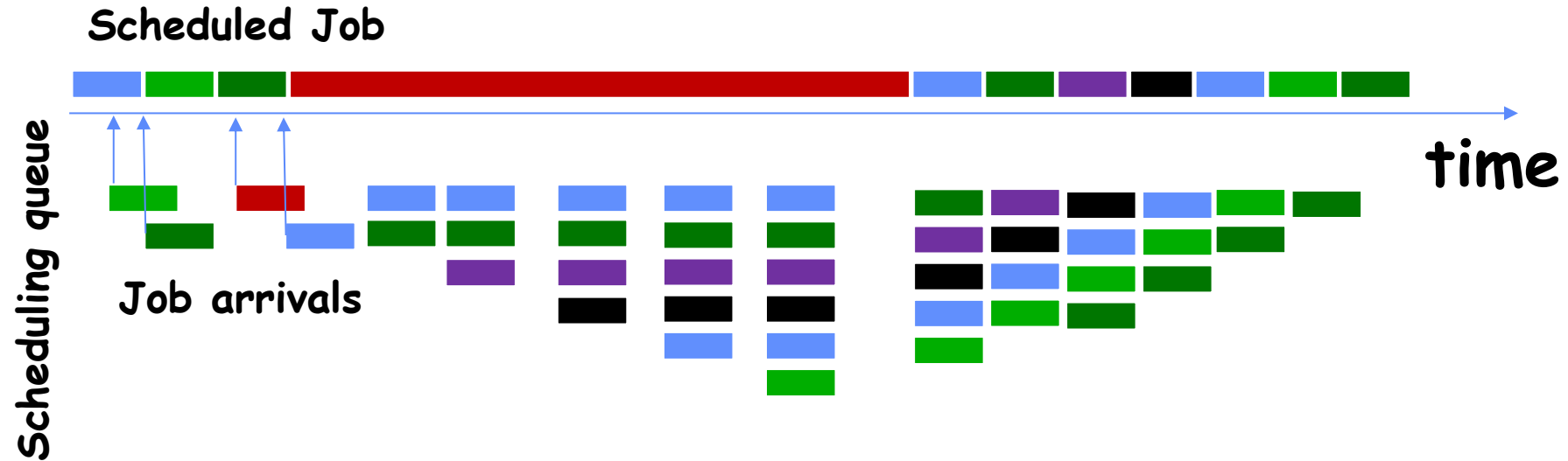
- Example continued:

- Suppose jobs arrive in the order: P_2 , P_3 , P_1 at time 0:



- Initial waiting times: P_1 : 6; P_2 : 0; P_3 : 3
- Response times: P_1 : 30; P_2 : 3; P_3 : 6
- Average initial waiting time: $(6 + 0 + 3)/3 = 3$ (vs. 17 before)
- Average response time: $(30 + 3 + 6)/3 = 13$ (vs. 27 before)

Convoy Effect



- With FCFS non-preemptive scheduling, convoys of small tasks tend to build up when a large one is running.



Long job



Short job



Short job

Round Robin (RR) Scheduling

- Round Robin Scheme:
 - Each job gets a small unit of CPU time (*time slice or time quantum*), usually 10-100 milliseconds
 - When quantum expires, the job is preempted and added to the end of the ready queue
 - If the current CPU burst finishes before quantum expires, the job blocks for IO and is added to the end of the ready queue
 - n jobs in ready queue and time quantum is $q \Rightarrow$
 - » Each job gets (roughly) $1/n$ of the CPU time
 - » In chunks of at most q time units
 - » No job waits more than $(n-1)q$ time units
- OS implementation:
 - Use a periodic timer interrupt to preempt the running job every time quantum, and send it to the back of the ready queue

Example of RR with Time Quantum = 20

- Example:

<u>job</u>	<u>Burst Time</u>
P_1	53
P_2	8
P_3	68
P_4	24

 - Suppose jobs arrive in the order of P_1, P_2, P_3, P_4 at time 0. Gantt chart:

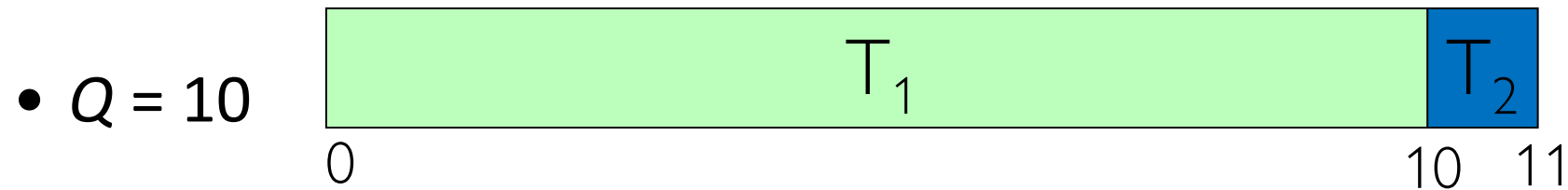
P_1	P_2	P_3	P_4	P_1	P_3	P_4	P_1	P_3	P_3	
0	20	28	48	68	88	108	112	125	145	153
 - Waiting times:
 $P_1 = (68-20) + (112-88) = 72$
 $P_2 = (20-0) = 20$
 $P_3 = (28-0) + (88-48) + (125-108) = 85$
 $P_4 = (48-0) + (108-68) = 88$
 - Average waiting time = $(72+20+85+88)/4 = 66\frac{1}{4}$
 - Response times: $P_1: 125; P_2: 28; P_3: 153; P_4: 112$
 - Average response time = $(125+28+153+112)/4 = 104\frac{1}{2}$
- Round-Robin scheduling
 - Pro: Better for short jobs, Fair
 - Con: Context-switching overhead adds up for long jobs

Quantum size

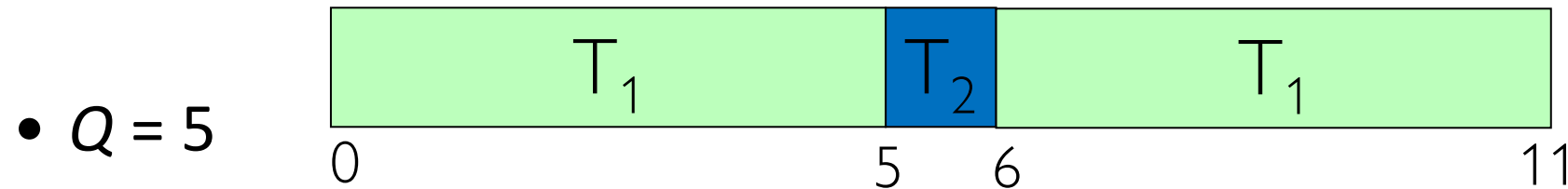
- Choice of time quantum size q :
 - q must be large with respect to context-switching overhead,
 - q too large: fairness is reduced. RR with infinite time quantum is equivalent to FCFS
 - q too small: too many context-switches with high overhead
- Typical time quantum in modern OSes is between **10ms – 100ms**
- Typical context-switching overhead is **0.1ms – 1ms**
 - Roughly **1%** overhead due to context-switching

Decrease Response Time w. Decreasing Quantum

- T_1 : Burst Length 10
- T_2 : Burst Length 1
- Suppose jobs arrive in the order of T_1 , T_2 at time 0



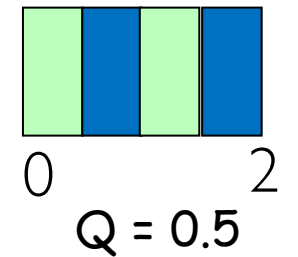
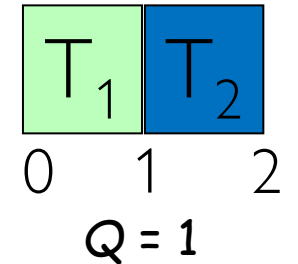
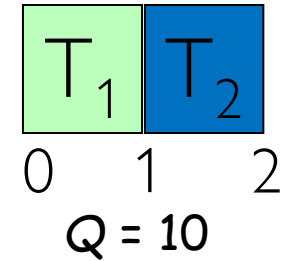
– Average Response Time = $(10 + 11)/2 = 10.5$



– Average Response Time = $(11 + 6)/2 = 8.5$

Response Time vs. Time Quantum

- T_1 : Burst Length 1
- T_2 : Burst Length 1
- Suppose jobs arrive in the order of T_1, T_2 at time 0
- $Q = 10$
 - Average Response Time = $(1 + 2)/2 = 1.5$
- $Q = 1$
 - Average Response Time = $(1 + 2)/2 = 1.5$
- $Q = 0.5$
 - Average Response Time = $(1.5 + 2)/2 = 1.75$



FCFS vs. Round Robin

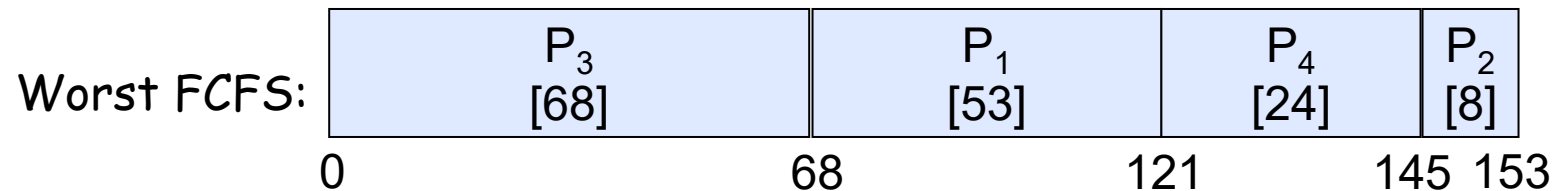
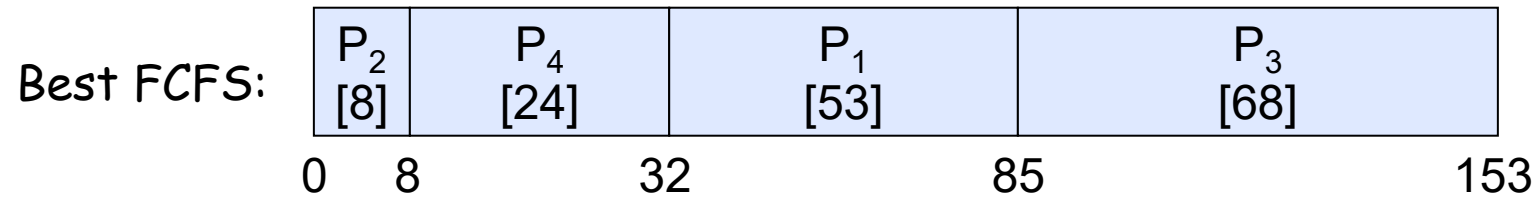
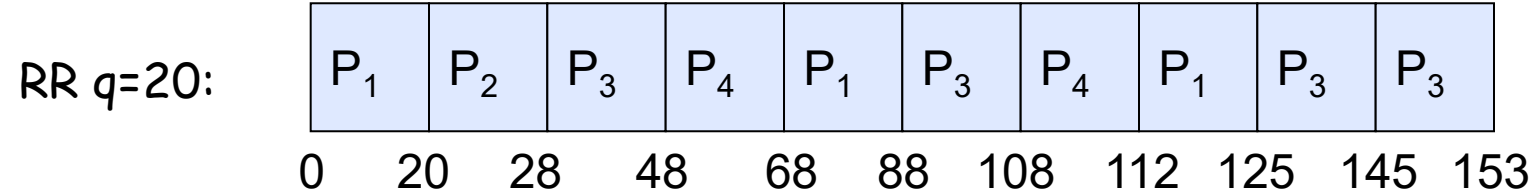
- Assuming zero-cost context-switching time, RR may not be better than FCFS, e.g., when all jobs have equal execution time
- Simple example: 10 jobs, each take 100s of CPU time
RR scheduler quantum of 1s
All jobs start at the same time
- response times:

Job #	FIFO	RR
1	100	991
2	200	992
...	...	...
9	900	999
10	1000	1000

- Both RR and FCFS finish at the same time
- Average response time is much worse under RR than FCFS
- Frequent context switches under RR hurts cache locality and increases job execution time due to increased cache miss rate

Consider the Previous Example

<u>Job</u>	<u>Burst Time</u>
P_1	53
P_2	8
P_3	68
P_4	24



- When jobs have uneven length, it seems to be a good idea to run short jobs first!

Earlier Example with Different Time Quantum

	Quantum	P ₁	P ₂	P ₃	P ₄	Average
Wait Time	Best FCFS	32	0	85	8	31¼
	Q = 1	84	22	85	57	62
	Q = 5	82	20	85	58	61¼
	Q = 8	80	8	85	56	57¼
	Q = 10	82	10	85	68	61¼
	Q = 20	72	20	85	88	66¼
	Worst FCFS	68	145	0	121	83½
Completion Time	Best FCFS	85	8	153	32	69½
	Q = 1	137	30	153	81	100½
	Q = 5	135	28	153	82	99½
	Q = 8	133	16	153	80	95½
	Q = 10	135	18	153	92	99½
	Q = 20	125	28	153	112	104½
	Worst FCFS	121	153	68	145	121¾

Average Wait Time and Response Time (Completion Time) may increase or decrease with varying time quantum

SJF and SRTF

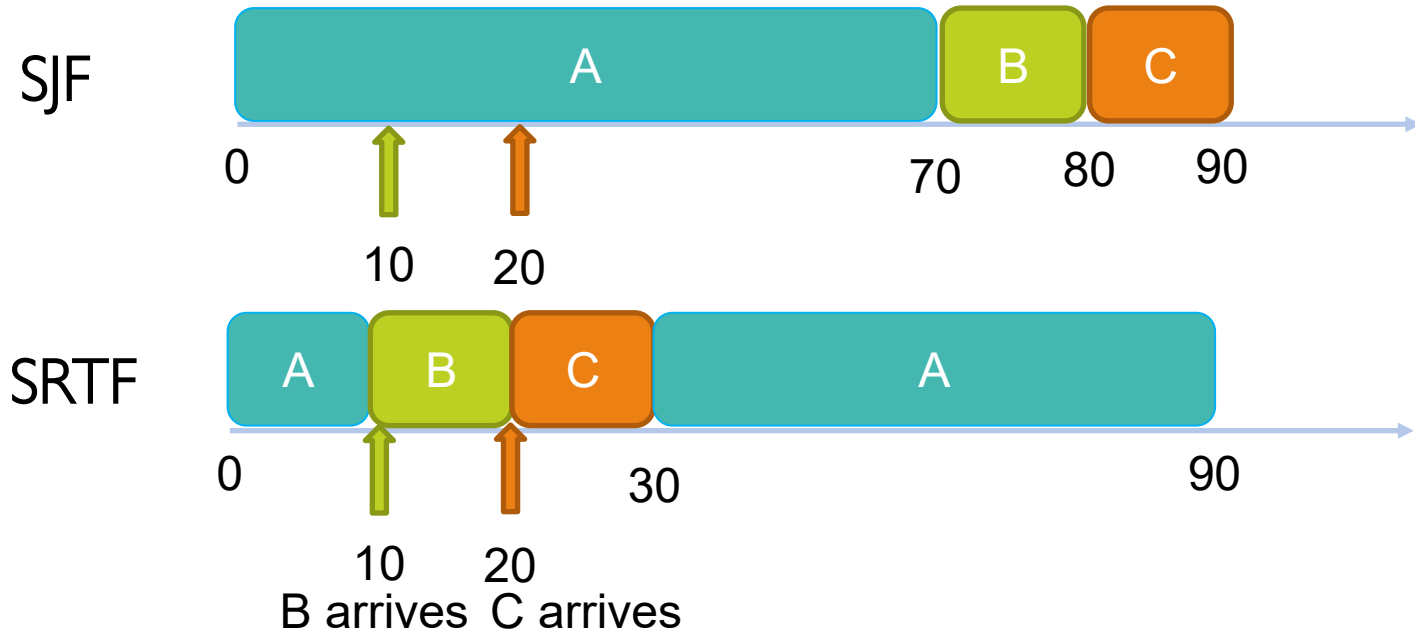
- If we know job execution times at arrival time (predict the future), then we can implement SJF and SRTF
- Shortest Job First (SJF):
 - Non-preemptive scheduling: Run whatever job has least amount of computation to do
 - Still suffers from convoy effect due to non-preemption
- Shortest Remaining Time First (SRTF):
 - Preemptive scheduling: if a new job arrives with remaining time less than remaining time of currently-executing job, preempt the current job
 - » In case of a tie (a new job arrives with remaining time equal to remaining time of currently-executing job), then do not preempt the current job
- Key idea: Give higher priority to short jobs and finish them quickly
 - Big benefit for short jobs, only small delay effect on long ones
 - Result is better average response time



SJF and SRTF Example

- SRTF achieves shorter average response time (Avg RT) than SJF, thanks to preemptive scheduling

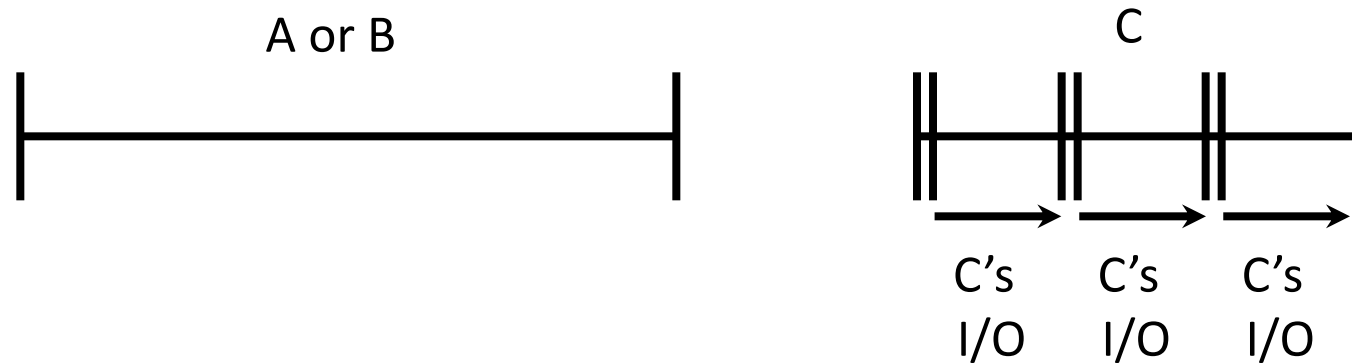
J o b	Arrival time	Exec Time	SJF Finishing Time	SJF Response Time	SRTF Finishing Time	SRTF Response Time
A	0	70	70	70	90	90
B	10	10	80	70	20	10
C	20	10	90	70	30	10
			Avg RT 70		Avg RT 37	



Optimality of SJF and SRTF

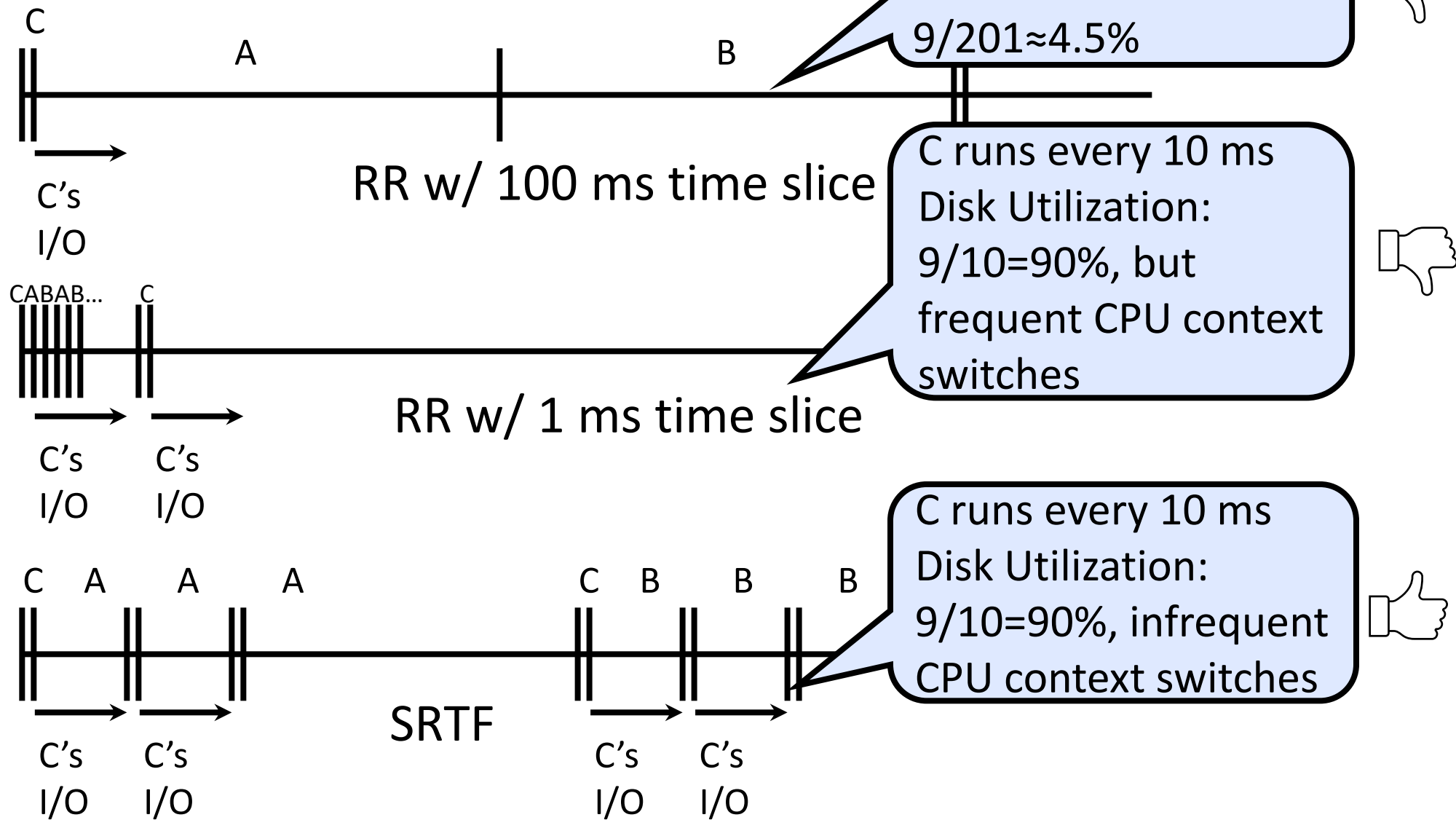
- SJF is the optimal scheduling algorithm for minimizing the average response time under the following assumptions:
 - All jobs only use the CPU (no I/O)
 - All jobs arrive at the same time
 - Job execution times are known in advance
 - Non-preemptive scheduling
- SRTF is the optimal scheduling algorithm for minimizing the average response time under the following assumptions:
 - All jobs only use the CPU (no I/O)
 - Job execution times are known in advance
 - Preemptive scheduling
- Comparison of SRTF with FCFS
 - If all jobs have the same length (execution time)
 - » SRTF becomes the same as FCFS (i.e. FCFS is optimal if all jobs the same length)
 - If jobs have varying length
 - » SRTF is better, since short jobs are not stuck behind long ones

Example to illustrate benefits of SRTF



- Three jobs:
 - A, B: both CPU bound, run for an hour
 - C: I/O bound, runs in a loop of 1ms CPU followed by 9ms disk I/O
 - If each job runs alone without interference, then C uses 90% of disk, A or B uses 100% of CPU
- With FCFS:
 - A and B may arrive and keep CPU busy for two weeks before C is scheduled
- What about RR or SRTF?

SRTF Example continued:



SRTF Discussions

- How to predict job execution time?
 - Runtime measurement and profiling for typical inputs
 - Offline static analysis
 - Difficult and error-prone in general
- Unfair
 - SRTF can lead to starvation if many small jobs arrive so large jobs never get to run
- SRTF Pros & Cons
 - Pros: Optimal in minimizing average response time)
 - Cons: Hard to predict job execution time; Unfair

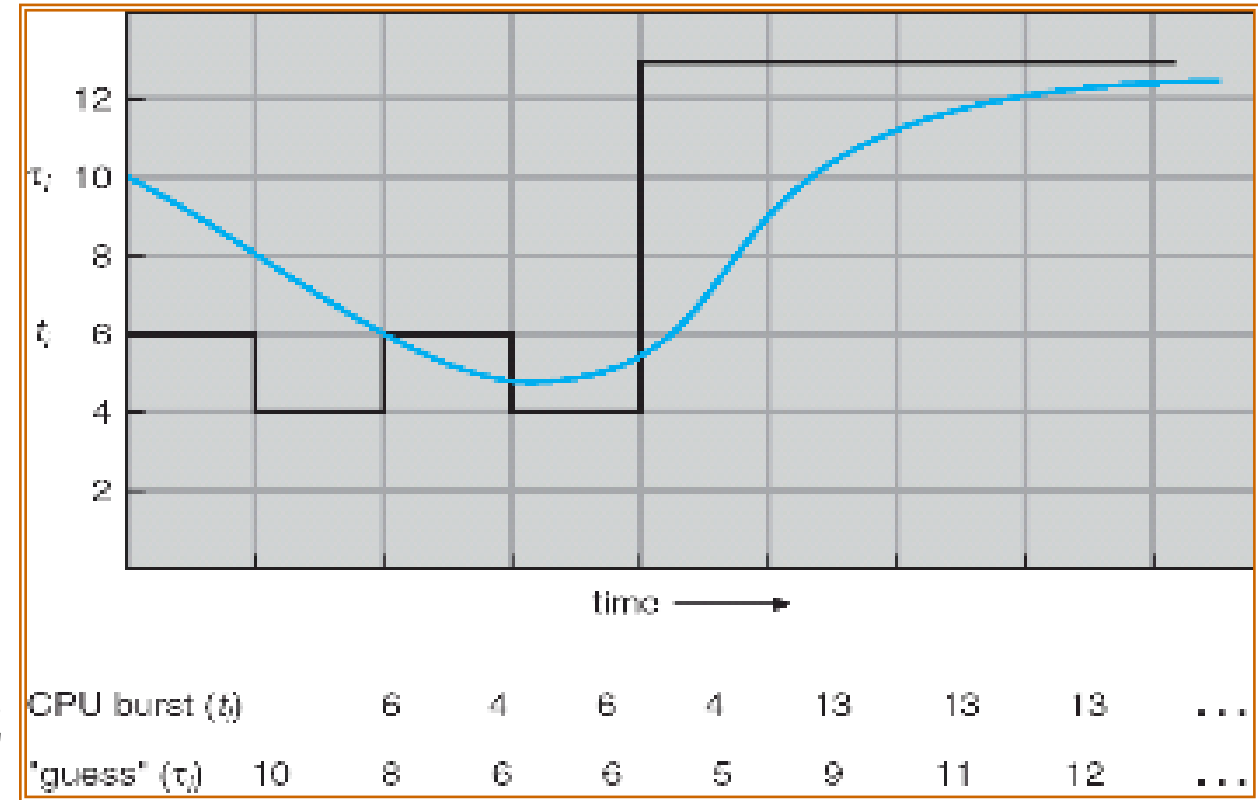


Predicting Length of the Next CPU Burst

- **Adaptive:** Changing policy based on past behavior
 - Works because programs have predictable behavior
 - » If program was I/O bound in recent past, it is likely to be I/O bound in future
- We can use **exponential moving averaging** $t_n = \alpha x_n + (1 - \alpha)t_{n-1}$, where:
 - x_n is the new input data point
 - t_{n-1} is the previous exponential moving average
 - α is the smoothing factor ($0 < \alpha < 1$)
 - α large: fast update of τ_n based on new input.
 - α small: slow update of τ_n based on new input.

Predicting Length of the Next CPU Burst: $\alpha=0.5$

- Compute $t_n = \alpha x_n + (1 - \alpha)t_{n-1}$ with initial guess $\tau_0 = 10$, assuming $\alpha=0.5$
- $t_1 = \alpha x_1 + (1 - \alpha)t_0 = 0.5*6 + 0.5*10 = 8$
- $t_2 = \alpha x_2 + (1 - \alpha)t_1 = 0.5*4 + 0.5*8 = 6$
- $t_3 = \alpha x_3 + (1 - \alpha)t_2 = 0.5*6 + 0.5*6 = 6$
- $t_4 = \alpha x_4 + (1 - \alpha)t_3 = 0.5*4 + 0.5*6 = 5$
- $t_5 = \alpha x_5 + (1 - \alpha)t_4 = 0.5*13 + 0.5*5 = 9$
- $t_6 = \alpha x_6 + (1 - \alpha)t_5 = 0.5*13 + 0.5*9 = 11$
- $t_7 = \alpha x_7 + (1 - \alpha)t_6 = 0.5*13 + 0.5*11 = 12$



Time Series 101: Exponential Moving Average, A Visual Guide
<https://www.youtube.com/watch?v=joHKNtPYtLo>

Predicting the Length of the Next CPU Burst: $\alpha=0.1$ or 0.9

- Compute $t_n = \alpha x_n + (1 - \alpha)t_{n-1}$ with initial guess $\tau_0 = 10$, assuming $\alpha=0.1$.
 - $t_1 = \alpha x_1 + (1 - \alpha)t_0 = 0.1*6 + 0.9*10 = 9.6$
 - $t_2 = \alpha x_2 + (1 - \alpha)t_1 = 0.1*4 + 0.9*9.6 = 9.0$
 - $t_3 = \alpha x_3 + (1 - \alpha)t_2 = 0.1*6 + 0.9*9.0 = 8.7$
 - $t_4 = \alpha x_4 + (1 - \alpha)t_3 = 0.1*4 + 0.9*8.7 = 8.3$
 - $t_5 = \alpha x_5 + (1 - \alpha)t_4 = 0.1*13 + 0.9*8.3 = 8.7$
 - $t_6 = \alpha x_6 + (1 - \alpha)t_5 = 0.1*13 + 0.9*8.7 = 9.2$
 - $t_7 = \alpha x_7 + (1 - \alpha)t_6 = 0.1*13 + 0.9*9.2 = 9.5$
- Compute $t_n = \alpha x_n + (1 - \alpha)t_{n-1}$ with initial guess $\tau_0 = 10$, assuming $\alpha=0.9$.
 - $t_1 = \alpha x_1 + (1 - \alpha)t_0 = 0.9*6 + 0.1*10 = 6.4$
 - $t_2 = \alpha x_2 + (1 - \alpha)t_1 = 0.9*4 + 0.1*6.4 = 4.2$
 - $t_3 = \alpha x_3 + (1 - \alpha)t_2 = 0.9*6 + 0.1*4.2 = 5.8$
 - $t_4 = \alpha x_4 + (1 - \alpha)t_3 = 0.9*4 + 0.1*5.8 = 4.2$
 - $t_5 = \alpha x_5 + (1 - \alpha)t_4 = 0.9*13 + 0.1*4.2 = 12.1$
 - $t_6 = \alpha x_6 + (1 - \alpha)t_5 = 0.9*13 + 0.1*12.1 = 13.0$
 - $t_7 = \alpha x_7 + (1 - \alpha)t_6 = 0.9*13 + 0.1*13.0 = 13.0$

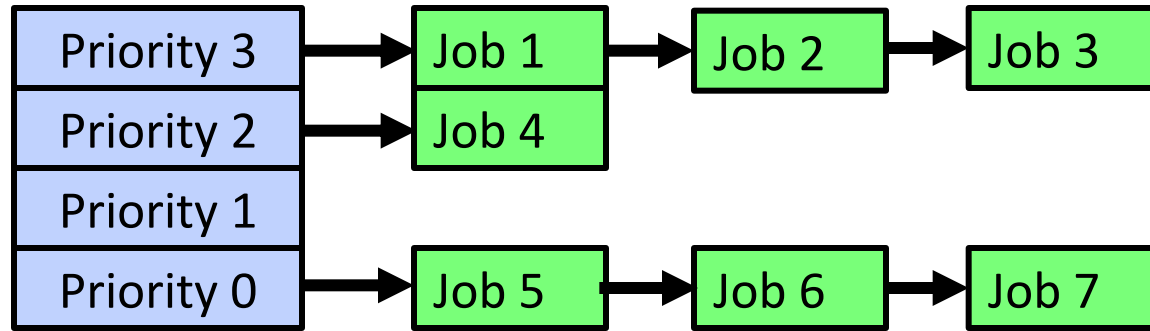
With low $\alpha = 0.1$, the EMA changes gradually and reacts slowly to new data, staying closer to the starting value.

With high $\alpha = 0.9$, the EMA responds quickly and closely tracks the latest data points.

Comparison Chart

Property	FCFS	SJF	SRTF	RR
Optimize Average Response Time		✓	✓	
Prevent Starvation	✓			✓
Prevent Convoy Effect			✓	✓
No Need to Predict Exec Time	✓			✓

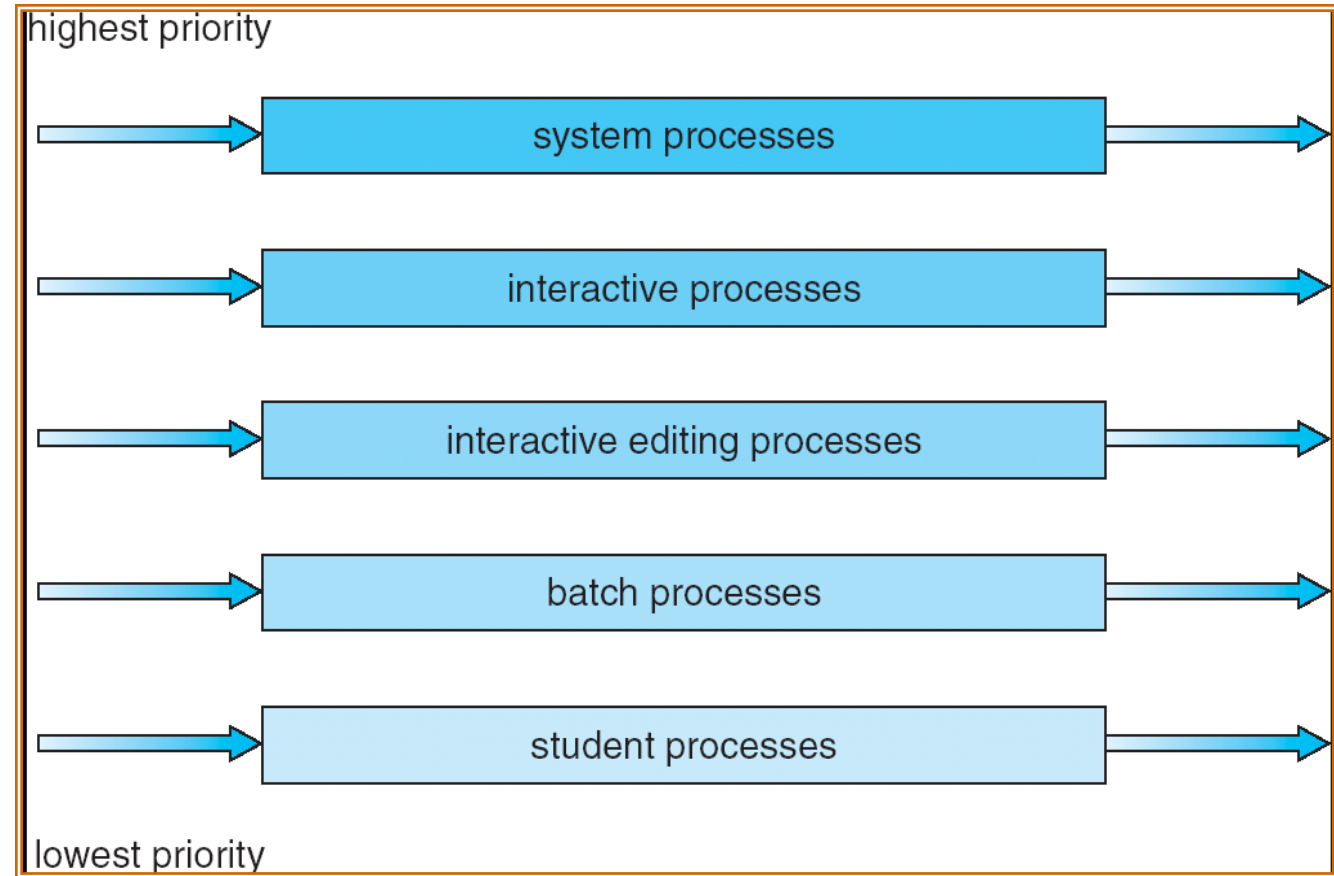
Fixed-Priority Scheduling



- Fixed-Priority Scheduling
 - Each job is assigned a fixed priority
 - Run the highest-priority job in the ready queue at any given time (may be preemptive or non-preemptive)
 - Jobs of equal priority are scheduled with RR
- SJF/SRTF are special cases of priority-based scheduling where priority is the predicted (remaining) job execution time
- Problem: starvation – low priority jobs may never execute
 - Sometimes this is the desired behavior!

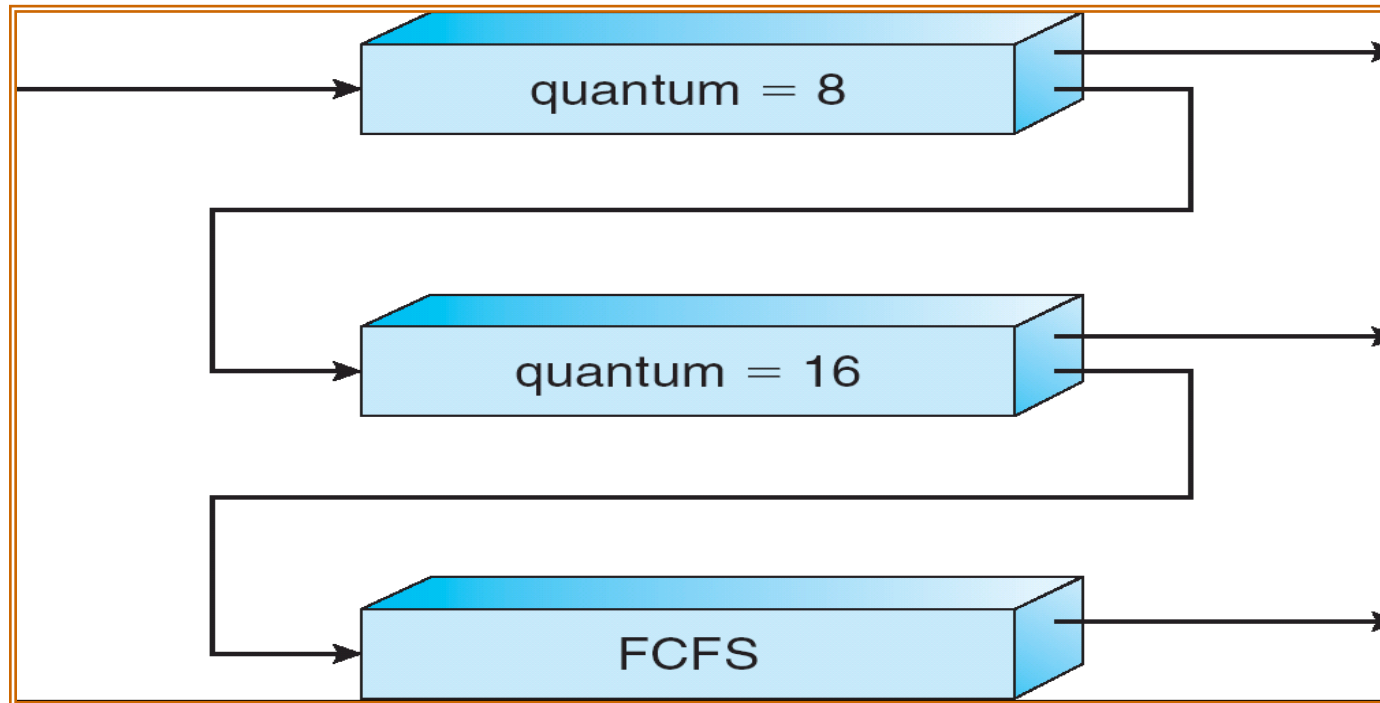
Multi-Level Queue Scheduling

- Ready queue is partitioned into multiple queues, each with different priority
 - Higher priority queues often considered “foreground” tasks
- Each queue has its own scheduling algorithm
 - e.g., foreground queue (interactive jobs/processes) with RR scheduling; background queue (batch jobs/processes) with FCFS scheduling
 - Typically time quantum increases with decreasing priority (highest: 1ms, next: 2ms, next: 4ms, etc)
- Scheduling between the queues
 - Fixed priority, e.g., serve all from foreground queue, then from background queue



Multi-Level Feedback Queue Scheduling

- Based on Multi-Level Queue Scheduling, but dynamically adjust each job's priority as follows:
 - It starts in highest-priority queue
 - If quantum expires before the CPU burst finishes, drop down one level
 - If it blocks for I/O before quantum expires, push up one level (or to top, depending on implementation)
 - Time quantum increases with decreasing priority, from 8, to 16, to infinity (RR with infinite time quantum is equivalent to FCFS)



Multi-Level Feedback Queue Scheduling Discussions

- MLFQ approximates SRTF:
 - Long-running CPU-bound jobs/processes are punished and drop down like a rock
 - Short-running I/O-bound processes are rewarded and stay near top
 - No need for prediction of job execution time; rely on past behavior to make decision
- User can game the scheduler:
 - e.g., put in a bunch of meaningless I/O like `printf()` to keep process in the high-priority queue
 - Of course, if everyone did this, this trick wouldn't work!

Conclusion

- **FCFS Scheduling:**
 - Run jobs in the order of arrival
 - Cons: Short jobs can get stuck behind long ones
- **Round-Robin Scheduling:**
 - Give each thread a small amount of CPU time when it executes; cycle between all ready threads
 - Pros: Better for short jobs
- **Shortest Job First (SJF)/Shortest Remaining Time First (SRTF):**
 - Run whatever job has the least execution time/least remaining execution time
 - Pros: Optimal (in terms of average response time)
 - Cons: Hard to predict execution time, Unfair
- **Priority-Based Scheduling**
 - Each job is assigned a fixed priority
- **Multi-Level Queue Scheduling**
 - Multiple queues of different priorities and scheduling algorithms
- **Multi-Level Feedback Queue Scheduling:**
 - Automatic promotion/demotion of jobs between queues to approximate SJF/SRTF